

TZANEEN-WESTFALIA ESTATE, South Africa

WMO: 681880

Lat: 23.7367S

Lon: 30.1127E

Elev: 896

StdP: 91.01

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.4	7.6	-3.4	3.2	20.1	-0.9	3.9	19.7	4.7	17.7	4.0	17.5	1.0	40	0.255

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.3	34.2	18.8	32.7	19.1	31.1	19.4	22.8	28.5	22.2	27.8	21.7	27.0	1.8	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	17.7	25.3	20.6	17.0	24.6	20.1	16.5	24.0	72.4	28.5	70.0	27.9	67.9	27.2	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.3	3.3	2.6	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.2	23.3	23.2	22.4	20.1	18.2	16.3	15.5	17.6	19.9	21.0	22.2	23.0
	DBStd	3.86	2.09	2.07	2.24	2.65	2.25	2.42	2.46	3.00	3.56	3.79	3.51	2.96
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	308	0	0	2	12	29	71	94	49	25	16	8	2
	CDD10.0	3727	412	368	383	304	254	188	171	237	297	342	367	403
	CDD18.3	994	154	136	128	66	25	9	6	28	72	100	125	147
	CDH23.3	6572	849	675	676	312	148	58	40	247	633	894	1004	1036
(13)	CDH26.7	2233	256	191	188	71	25	6	4	61	233	391	411	397
(14)	Wind	WSAvg	1.0	1.0	1.1	1.0	0.9	1.0	1.0	1.1	1.1	1.2	1.1	1.0
(15) Precipitation	PrecAvg	1045	186	232	153	48	19	17	16	11	19	54	127	163
	PrecMax	2517	403	1187	470	106	122	68	72	39	91	127	271	359
	PrecMin	507	41	30	56	5	0	0	2	1	1	4	29	63
	PrecStd	459	111	265	101	31	27	19	21	9	23	32	63	65
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.4	33.6	32.7	30.7	29.2	27.5	26.6	30.9	34.4	36.1	35.4	35.8
		MCWB	21.5	20.9	19.6	19.1	15.9	13.2	13.2	15.2	16.0	18.4	18.0	20.6
	2%	DB	31.4	30.6	30.5	28.8	26.8	24.9	24.0	28.5	31.8	33.5	33.9	32.8
		MCWB	20.9	21.4	20.1	19.3	16.2	13.2	13.1	15.0	16.0	17.8	18.6	20.5
	5%	DB	29.4	29.0	28.8	26.9	24.9	22.8	22.0	26.1	29.5	31.0	31.5	30.9
		MCWB	20.7	21.1	20.4	18.8	16.0	13.1	12.9	14.3	15.9	18.0	19.2	20.8
	10%	DB	27.6	27.4	27.1	25.0	23.1	21.0	20.1	23.8	27.2	28.3	29.0	28.8
		MCWB	20.6	20.8	20.3	18.5	15.7	13.2	12.8	13.8	15.8	17.9	19.5	20.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.7	23.2	23.0	21.8	18.9	16.5	15.6	18.6	19.5	21.6	22.4	23.3
		MCDB	29.8	28.9	28.1	26.8	24.3	22.3	21.4	24.4	27.3	28.7	30.2	29.2
	2%	WB	22.7	22.3	22.2	20.8	17.9	15.2	14.7	16.6	18.5	20.4	21.3	22.4
		MCDB	28.0	27.8	27.1	25.7	23.3	21.2	19.9	24.9	26.5	27.5	27.8	28.7
	5%	WB	21.9	21.7	21.5	20.1	17.2	14.4	13.9	15.6	17.6	19.6	20.6	21.7
		MCDB	26.9	26.9	26.1	24.7	22.4	19.9	19.1	22.5	25.4	26.6	26.6	28.1
	10%	WB	21.3	21.2	20.9	19.3	16.4	13.7	13.3	14.7	16.7	18.7	19.9	21.1
		MCDB	25.6	26.0	25.0	23.3	21.4	18.7	18.1	20.9	24.1	25.5	25.7	27.2
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	7.9	8.8	8.5	11.3	11.7	11.1	12.4	12.1	11.1	9.9	9.1
		MCDBR	12.0	11.3	12.4	12.3	14.4	14.4	13.9	16.4	16.9	16.8	15.0	13.6
		MCWBR	4.2	4.2	4.8	5.3	5.7	6.1	5.8	6.1	5.8	5.7	5.1	4.5
	5% WB	MCDBR	9.4	9.6	9.3	10.1	11.4	11.7	10.8	12.8	13.7	12.4	10.9	11.4
		MCWBR	3.9	4.0	4.4	4.7	5.6	5.8	5.3	5.7	5.5	5.2	4.5	4.4
(40) Clear-Sky Solar Irradiance	taub		0.388	0.384	0.375	0.356	0.323	0.308	0.307	0.358	0.416	0.432	0.401	0.397
	taud		2.401	2.413	2.429	2.439	2.484	2.505	2.496	2.340	2.210	2.214	2.338	2.372
	Ebn at Noon		956	945	926	900	893	888	901	884	872	890	937	948
	Edh at Noon		128	124	116	106	92	86	90	115	142	149	135	132
(44) All-Sky Solar Radiation	RadAvg		6.22	5.99	5.38	4.63	4.36	3.89	4.13	4.84	5.50	5.80	6.01	6.10
	RadStd		0.47	0.55	0.46	0.35	0.20	0.26	0.17	0.27	0.36	0.40	0.55	0.53

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page